

AVESH SHAH

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Controls and robotics engineer with 10+ years developing production-oriented vehicle control, state estimation, sensor fusion, and autonomy software. Experience spans NMPC/MPC, EKF/LQR/LQI, vehicle dynamics, ROS2, embedded platforms, GNSS, and ML-enabled automotive functions.

CORE SKILLS

Controls & Systems: Kalman Filter (EKF), Particle Filter, LQR/LQI, NMPC/MPC, RLS Estimation, Vehicle Dynamics, Sensor Fusion, Torque Control, Regen Braking, Feedback Control

Software & Platforms: MATLAB/Simulink, Python, C++, acados, CasADi, ROS2, Docker, LabVIEW, TFLite/YAMNet, JSON/YAML

Hardware & Protocols: NXP S32G, Raspberry Pi, DSpace, ETAS/INCA, CAN/SocketCAN, I2C/UART, GNSS/u-blox, MotoHawk

Process & Leadership: AUTOSAR, ASPICE, ISO 26262, DOORS/Polarion, Git/Azure, CI/CD, Agile/JIRA, OEM Workshops

PROFESSIONAL EXPERIENCE

Magna International - Troy, MI

Senior Software Engineer | Apr 2024 - Present

- Leading development and technical ownership of a constraint-aware nonlinear Model Predictive Control (NMPC) framework for real-time trajectory tracking and rollover-risk mitigation, using CasADi for symbolic model, cost, and constraint formulation and acados for real-time optimal control, with vehicle-dynamics and actuator constraints, coordinated differential braking, and bounded steering intervention.
- Supporting project leads with task tracking, test planning, and risk assessment inputs.
- Collaborating with ADAS and validation teams to review test results, debug issues, and update software baselines.
- Ensuring product deliverables meet technical and timing requirements across cross-functional reviews.
- Supported development of a gateway module spanning vehicle control and connectivity functions, including Wi-Fi and GNSS integration.
- Built EKF-based state estimation for Vx/Vy, heading, and pose fusing IMU, GPS, and wheel speeds; IMU calibration via I2C; roll/pitch via Kalman and Madgwick filters.
- Designed lateral control with rollover index / lateral-g limits and steering impulse; longitudinal acceleration/deceleration and regenerative-braking blending; range estimation.
- Integrated GNSS-based positioning with NTRIP corrections into ROS2/Docker pipelines for cm-level accuracy.
- Developed an ML-based algorithm to detect approaching emergency vehicles using YAMNet; built a ROS2/Docker CODEC interface for alerts and advertisements audio.
- Developed mass/grade estimation using Recursive Least Squares (RLS), achieving approximately 10% error margin on representative routes.

Dana Incorporated - Novi, MI

Technical Lead | Jan 2023 - Apr 2024

- Led a team of 6 engineers for software feature development for BEV transmission systems under AUTOSAR and ASPICE frameworks.
- Designed and implemented control logic for torque management, lubrication, and thermal handling features.
- Conducted internal technical reviews and supported feature validation and calibration activities.
- Contributed to project planning, task definition, and coordination with program management and OEM partners.

FEV North America - Auburn Hills, MI

Senior Controls Engineer | Oct 2018 - Jan 2023

- Developed torque path control algorithms including creep, dual-path shift energy management, e-coasting, One Pedal Drive, brake blending, and cost-based torque split optimization for hybrid and BEV programs.
- Designed LQR-based engine start/idle control profiling for faster actuation with minimal delay and regenerative-brake interfaces for smooth shift transitions.
- Led supplier workshops to align requirements, timelines, and performance validation criteria.
- Mentored junior engineers on Simulink modeling standards, documentation, and testing procedures.

Hybrid Design Services Inc. - Troy, MI

Controls Engineer | Dec 2016 - Oct 2018

- Developed hybrid/EV control strategies using MATLAB/Simulink and Stateflow.
- Built a prototype BMS with SOC estimation via EKF, power-limit calculation, and contactor control with weld-check logic.
- Developed a battery disconnect unit (BDU) watchdog capable of overriding the client BMS to protect the high-voltage battery pack.
- Implemented a supervisory controller to monitor and log faults via OBD-II and transmit logged data over FTP to a secure server.
- Performed third-party BMS calibration and HV battery-pack testing covering fuse current, isolation monitoring, contactor weld checks, and cell balancing.

General Motors - Pontiac, MI

R&D Test Systems Engineer | Aug 2014 - Dec 2016

- Developed Simulink/LabVIEW controls for gasoline and diesel research engines.
- Built FPGA/DSPACE test systems for prototype ECU validation and CAN communication.
- Performed test-cell commissioning by integrating Rapid Control Prototype ECUs and calibrating sensors including MAF, cam phaser, crank encoders, and thermocouples.

EDUCATION & CERTIFICATION

M.S., Automotive Engineering (GATE Fellow) - Clemson University | GPA 3.57

B.Tech., Mechanical Engineering - National Institute of Technology, Jaipur | GPA 8.37

Certification - Applied Machine Learning

SELECTED TECHNICAL PROJECTS

Constraint-Aware NMPC - Trajectory Tracking & Vehicle Stability

- Built an 11-state real-time nonlinear MPC controller for pose/trajectory tracking and rollover-risk mitigation, re-solving constrained optimization at each control step with acados/SQP and CasADi.
- Incorporated vehicle-dynamics and actuator limits, combined-slip/friction constraints, differential braking, bounded steering intervention, and closed-loop validation in CARLA.

77 GHz FMCW Radar & Adaptive Cruise Control Lab

- Developed a ROS2/Gazebo radar-processing pipeline spanning Range FFT, Doppler FFT, Range-Doppler mapping, angle estimation, CFAR detection, tracking, and ACC target selection.
- Modeled a 77 GHz FMCW sensor with 64-chirp frames and 4 RX channels to estimate target range, relative velocity, and direction for closed-loop vehicle-following experiments.
- Integrated tracked radar detections with path-aware lead selection and adaptive cruise control using relative radial velocity, constant-time-headway gap control, and predicted-path corridor projection to reject adjacent-lane targets on curves.

Vx/Vy EKF with Slip-Aware Sensor Fusion

- Implemented a nine-state EKF fusing wheel speeds, IMU yaw-rate/acceleration, and GPS velocities, with slip-aware noise scaling, low-speed yaw handling, delayed-measurement buffer replay, and bias estimation.

PointPerfect NTRIP Client (ROS2)

- Built a TLS NTRIP client that discovers mountpoints, sends live GGA, streams SPARTN/RTCM3 corrections to u-blox receivers, auto-reconnects, and integrates with ROS2/Docker for cm-level positioning.

Emergency Vehicle Detection (Audio ML)

- Developed a real-time binary siren detector using YAMNet embeddings and a custom TensorFlow Lite head, publishing detection status through ROS2 and integrating passenger-audio alert duck/restore behavior.